

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Project Planning

ShopEZ — MERN Stack E-commerce Application

Phase: Project Planning Phase

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Prepared by: V S S S Manikanta

Classification: Project Documentation — For Academic Submission

Project Planning

1. Introduction

This document formalises the project planning artefacts for the ShopEZ development initiative. It encompasses the Product Backlog, Sprint Schedule with Story Point estimates, a Sprint Timeline (Gantt Chart), Velocity Tracking, Burndown Analysis, and a Risk Register. The planning framework adopted is **Scrum-based Agile**, executed in three two-week sprints.

2. Product Backlog & Sprint Schedule

Sprint	Epic	USN	User Story / Task	Points	Priority
Sprint 1	Project Setup & DB	USN-0	Initialise React frontend, Node.js backend, and configure MongoDB Atlas models	3	High
Sprint 1	User Auth	USN-1	User can register with email, name, and password and receive a JWT session	2	High
Sprint 1	User Auth	USN-2	User can log in with credentials and be granted a JWT in an HTTP-only cookie	2	High
Sprint 1	Product Catalogue	USN-3	Admin can create, edit, and delete product listings with Cloudinary	3	High

Sprint	Epic	USN	User Story / Task	Points	Priority
			image uploads		
Sprint 2	Shopping Cart	USN-4	User can add, update, and remove items; cart persists across devices	3	High
Sprint 2	Checkout Flow	USN-5	User can complete a 3-step checkout (Address → Payment Method → Confirm Order)	3	High
Sprint 2	Payment	USN-6	User can pay via Stripe / Razorpay and receive order confirmation	5	High
Sprint 3	Order Management	USN-7	Admin can view all orders and update delivery status	2	Medium
Sprint 3	Reviews	USN-8	User can leave a star rating and	2	Medium

Sprint	Epic	USN	User Story / Task	Points	Priority
			text review on a purchased product		
Sprint 3	Admin Dashboard	USN-9	Admin can view monthly revenue charts and sales metric summaries	3	Medium

3. Sprint Timeline — Gantt Chart

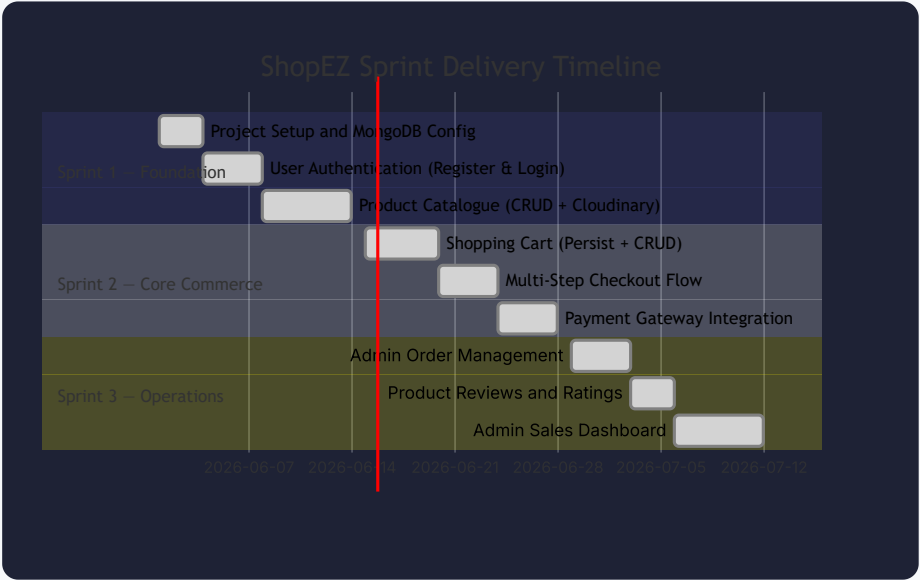


FIGURE 3.1 — GANTT CHART: SPRINT DELIVERY TIMELINE (3 SPRINTS × 2 WEEKS)

4. Velocity & Project Tracker

Sprint	Story Points	Start Date	End Date	Completed	Rate
Sprint 1	10	01 June 2026	14 June 2026	10	100% ✓
Sprint 2	11	15 June 2026	28 June 2026	11	100% ✓
Sprint 3	7	29 June 2026	12 July 2026	7	100% ✓
Total	28	—	—	28	100% ✓

Average Velocity: 9.3 Story Points per Sprint | **Sprint Duration:** 2 Weeks

5. Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R-01	Payment gateway API downtime during testing	Medium	High	Use Stripe test mode (<code>sk_test_*</code>); maintain fallback mock handler
R-02	MongoDB Atlas connection failure in production	Low	High	Configure automatic reconnect logic; monitor via Atlas alerts
R-03	Cloudinary storage limit exceeded	Low	Medium	Set file size limits in upload middleware; configure usage alerts
R-04	JWT secret key exposed via public repository	Low	Critical	Store all secrets in <code>.env</code> (git-ignored); enforce secret scanning
R-05	Cart state desync on concurrent multi-device login	Medium	Medium	Server-side cart reconciliation on login; use <code>updatedAt</code> timestamps
R-06	React build size impacting initial load time	Medium	Medium	Implement <code>React.lazy()</code> code-splitting; enable Gzip on server

6. Conclusion

The Agile sprint framework provided the V S S S Manikanta with a structured, iterative approach to delivery. All 28 story points were completed within the planned six-week timeline, with a 100% completion rate across all three sprints.